

Q. (i) $L_i \rightarrow$ Hydration energy $\uparrow$ $\left[\frac{H^+}{I} \frac{e^4}{r^2} \right]$ $\rightarrow$ change density.

(ii) Properties depends on e^- & their arrangement.

* S-block $\rightarrow$ with $HCl \rightarrow$ give Cl^- form Test $\rightarrow$ more volatile.

$\hookrightarrow$ with H_2SO_4 & $HNO_3 \rightarrow$ not give $\rightarrow$ because the SO_4^{2-} & NO_3^- salts are less volatile & even decompose on flame giving oxides.

Q. (i).

~~ANAL.~~ ANAL. $NaOH + HCl \rightarrow$ Phenolphthalein / methyl orange. $KMnO_4$ (self indicator) $K_2Cr_2O_7$ (self indicator) $EDTA$ vs $Zn(2) \rightarrow$ xylenol orange or phenolphthalein, $AcOH$ vs $NaOH \rightarrow$ phenolphthalein.

LANBERT LAW: It is the linear relationship b/w Absorbance & concentration of absorbing species. It is combination of 2 laws $\rightarrow$ Beer's law, Lambert law.

Mathematical Derivation. $\rightarrow$ Measuring the absorbance of solⁿ.



Absorbance $= A = \frac{I_0 - I}{I_0}$

Transmission $= T = \frac{I}{I_0}$

CASE I

$\Rightarrow$ Non Absorbing Sample

$\Rightarrow$ Transparent.

$I = I_0$

$T = 100\%$

Absorbance A ; $A = \log_{10} \left(\frac{I_0}{I} \right)$

CASE II

$\Rightarrow$ Absorbing sample

$\Rightarrow$ opaque sample

$I = 0$

$T = 0\%$

LANBERT'S LAW.

Absorbance of light in homogeneous sol. is directly proportional to the length of sample in which the light pass.

$A = \log_{10} \left(\frac{I_0}{I} \right) \propto l$

ϵ = Molar extinction coeff.

$A = \epsilon c l$

absorption intensity.

l = Path the light has travelled

BEER'S LAW.

Absorbance of light in sample/sol. is directly proportional to conc. of sol in which light travel.

$A = \log_{10} \left(\frac{I_0}{I} \right) \propto c$

$A = \epsilon c l$

c = conc. of sample solⁿ.

Combining both laws.

$A = \log_{10} \left(\frac{I_0}{I} \right) = \epsilon c l$

$A = \epsilon c l$

c = conc. l = length,

* Beer-L. Law fails at higher concn because the linearity of the law is limited to chemical factors, when concn is high, the refractive index changes due to light deviate

★ **Auxochrome** $\Rightarrow$ is a functional group of atoms attached to chromophore which modifies the ability of the chromophore to absorb light & change intensity of the absorption. Ex $\rightarrow$ $-\text{OH}$, $-\text{NH}_2$, $-\text{CH}_2$, etc.

★ **Chromophore** is an unsaturated group that absorbs light & reflect it at specific angle to give hue. Ex $\rightarrow$ Nitro group, azo compounds.

★ Different types of shifts noted in UV-vis spectrum?

(1). **Bathochromic shift** $\Rightarrow$ It is a shift of peak position (I_{max}) to a higher wavelength due to effect of substituent group. It is also known as red shift.

(2). **Hypsochromic shift** $\Rightarrow$ It is a shift of I_{max} to a lower wavelength also known as blue shift.

(3). **Hyperchromic shift** $\Rightarrow$ Due to increase in intensity of absorption maxima or due to introduction of auxochrome. Ex $\rightarrow$ pyridine & 2-methyl pyridine.

(4). **Hypochromic shift** $\Rightarrow$ Decrease in ϵ .

TGA

TGA is performed by ~~changing~~ raising temp. of a sample as its weight is measured on an analytical balance. In TGA, mass loss is observed if a thermal event involves loss of a volatile component.

INSTRUMENTATION

The sample is placed in an analytical balance. The restoring force is directly proportional to change in weight.

Application $\Rightarrow$ (a). Determination of both primary & secondary standard in titration. purity & chemical stability can be calculated.

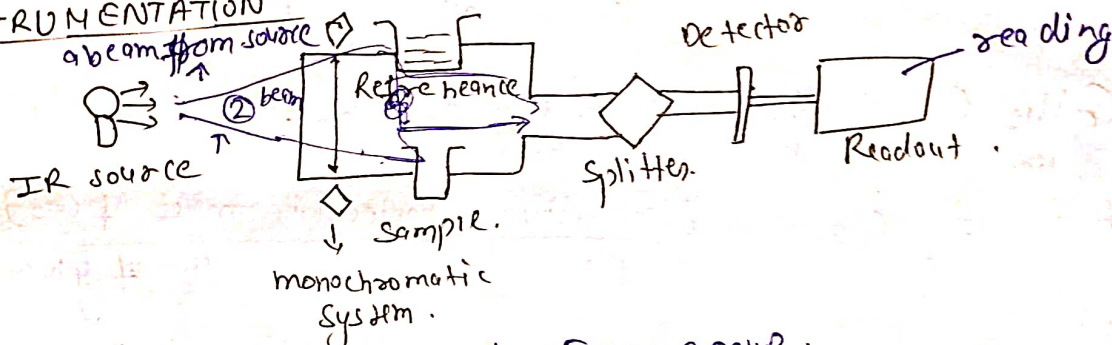
(b). Thermal stability of polymers can be calculated.
 (c). determination of composition of alloys & mixtures.
 (d). studying oxidation of alloys using the TGA curve.

IR spectroscopy. (Infrared region).

Infrared spect. is the measurement of the interaction of Infrared radiation with matter by absorption, emission, or reflection study.

It is used to identify ~~the~~ chemical substances ~~that~~ functional groups in solid, liquid or gaseous form.

INSTRUMENTATION



Application :- Identification of func-group.
 Studying progress of rxn.
 Detection of impurities.
 Quantitative analysis.

$$\textcircled{10}. f_1 = \frac{1}{2\pi} \sqrt{\frac{k}{36}} \quad \& \quad f_2 = \frac{1}{2\pi} \sqrt{\frac{k}{32}}$$

$$\frac{f_2}{f_1} = \sqrt{\frac{36}{32}} = \frac{3}{2\sqrt{2}}$$

$$\Rightarrow \quad \pi = f_2 = 2940 \cdot 14$$

$$\textcircled{11}. \text{CO}_2 \Rightarrow N=3$$

$$\text{SO}_2 \Rightarrow N=3$$

$$\text{CH}_4 \Rightarrow N=5$$

$$\text{H}_2\text{O} \Rightarrow N=3$$

$$R.D = 2$$

$$R.D = 3$$

$$R.D = 3$$

$$R.D = 3$$

$$V.D = 3N - 5 = 4$$

$$V.D = 3N - 6 = 3$$

$$V.D = 3N - 6 = 9$$

$$V.D = 3N - 6 = 3$$

Ans. 14. 6 Principles of Green chemistry

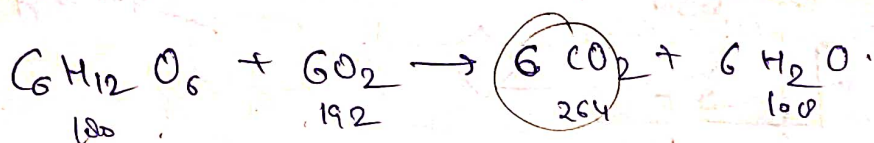
- (1). Prevent waste :- Design chemical syntheses to prevent waste.
- (2). Atom economy :- Reduce waste at molecular level by no. of atoms.
- (3). Design safer chemicals :- Minimise toxicity directly by molecular design.
- (4). Catalysis :- Use catalyst in order to increase selectivity, minimise wastage.
- (5). Energy efficiency :- Energy requirements should be minimised.

(6). Renewable feed stocks $\Rightarrow$ Raw material should use, where practicable, be renewable.

(7) Reduce d

Ans. 15. Atom economy is a measure of amount of starting materials that end up as useful products.

$$\text{Atom economy} = \frac{\text{Mass of desired output}}{\text{Mass of all products}} \times 100.$$



$$\begin{aligned} \text{A.E. of CO}_2 &= \frac{[\text{relative molecular mass of CO}_2]}{[\text{Sum of relative molecular masses of glucose \& O}_2]} \times 100 \\ &= \frac{264}{372} \times 100 = 70.96. \end{aligned}$$

Ans. 16. Green solvents are environmental friendly solvents, or biosolvents, which are derived from the processing of agricultural crops.

Ex $\rightarrow$ Ethyl lactate, ethanol, acetone, etc.

Ans. 17. (a). Solvent free rxn $\Rightarrow$ A dry rxn or solid-state rxn or solventless rxn is a chemical rxn takes place in absence of a solvent.

$\rightarrow$ It follows the fifth principle of G.C which avoids using toxic solvents in chemical rxn.

Ex $\rightarrow$ Oxidative coupling Rxn of phenols with FeCl_3 .

(b). Microwave assisted rxns $\Rightarrow$ is a unique & simple technique which offer fast & efficient processing of materials with higher reproducibility. (productivity),
Ex $\rightarrow$ rxn of arylboronic acid with aryl aldehydes in presence of diethylzinc & arisidine.

(c). Ultrasound-assisted rxns: $\rightarrow$ Ultrasound enhances the reactivity of molecules towards many chemical reactions via the process of acoustic cavitation.

Ans. 18. Water Treatment plant: $\rightarrow$ 3 main stages $\begin{matrix} \swarrow \text{Primary} \\ \rightarrow \text{Secondary} \\ \searrow \text{Tertiary} \end{matrix}$ to bottom

Primary: $\rightarrow$ Waste water is held in a settling tank where heavier solids settle down & while lighter solids float to surface & then discharge to secondary.

Secondary: $\rightarrow$ Degrade biological content of waste through aerobic biological processes.

Tertiary: $\rightarrow$ main aim to raise quality of water. It removes pathogens, which ensure that water is safe for drinking purposes.

Ans. 19. TDS (Total dissolved solids) ~~is~~ is a measure of dissolved content of inorganic, organic substances. It is measured in ppm. TDS is study water quality & used as indicator for drinking water.

BOD (Biochemical oxygen demand) $\rightarrow$ represent amount of oxygen consumed by bacteria & presence of a sufficient concn of dissolved oxygen is essential for aquatic life.

TOC (Total organic carbon) is measure of total amount of carbon in O.C in pure water & aqueous systems. When completing T.O.C analysis, the following is measured:-

- TC - Total Carbon
- TIC - Total Inorganic Carbon.
- POC - Purgeable Organic Carbon.
- NPOC - Non " " "
- DOC - Dissolved Organic Carbon.
- NDOC - Non " " " " "

Ans. 20.

Eutrophication is a process in which a water body becomes overly enriched with nutrients, leading to plentiful growth of simple plant life.

This is a major issue because it reduces the oxygen dissolved in water which affects aquatic life. after some times Eutrophic water becomes "dead zones" that are incapable of supporting life.

Ans. 21. 2 methods for water disinfection.

(1) Distillation: Water is purified by distillation for chemical purposes. Water is placed in a flask & is heated after that the vapours condense after passing through condenser & collected pure water is collected in other flask.

(2) Treatment of water: This process involves sedimentation of bigger particles at bottom of tank & then chlorination of filtered water.

Ans. 22.

HARDNESS.

Total Hardness is ~~or~~ ^{can be equivalent} to total calcium & magnesium hardness.

Temporary Hardness: due to $\text{Ca}(\text{HCO}_3)_2$ & $\text{Mg}(\text{HCO}_3)_2$ are converted into the corresponding insoluble carbonates & are precipitated.

Permanent hardness: due to Cl^- & SO_4^{2-} of Mg & Ca. which will not be precipitated by boiling.

Hardness is measured in units of ppm.

$$\star \Lambda_m = KV$$

$$\star K = \frac{G}{R}$$

$$KR = G$$

$$\therefore [K_1 R_1 = R_2 K_2]$$

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

$$\star \Lambda_m = \frac{K \times 1000}{M}$$

$$\star \Lambda_m = \Lambda_m^\circ - \frac{A\sqrt{C}}$$

$$\star E = E^\circ - \frac{0.0591}{n} \log Q$$

$$\star \alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

Ans. 48. Molar conductivity $\Rightarrow$ of a solⁿ at a given concⁿ is conductance of vol. V of a solⁿ containing 1 mole of the electrolyte kept b/w 2 electrodes with area of cross A & distance of unit length.

$$\Lambda_m = KA$$

M.C $\uparrow$ with $\downarrow$ in C . This is becoz the total vol. V of solⁿ containing one mole of electrolyte increases on dilution.

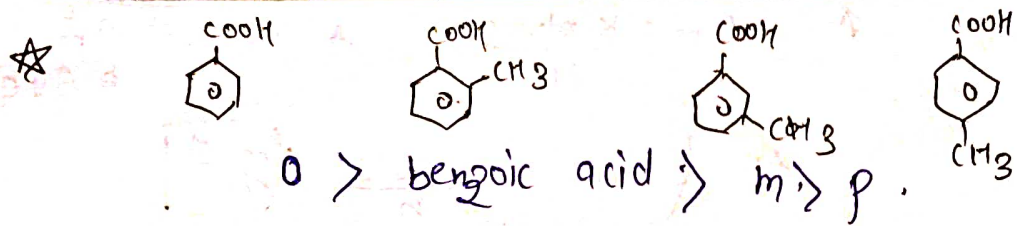
Ans. 49. (i). Intermediate compound formation theory. Compounds are formed with less energy consumption than needed for actual rxn. The intermediate compound being unstable combines with another reactant to form desired product & catalyst is regenerated. Usually, Homogeneous catalysis follow I.C.F.T. of catalysis.

Ans. 50. An enzyme is a substance that acts as a catalyst in living organisms, regulating rate of chemical reaction without itself being used.

Starch $\xrightarrow{\text{Diastase}}$ Maltose $\xrightarrow{\text{Maltase}}$ Glucose.

Urea $\xrightarrow{\text{urease}}$ $\text{NH}_4 + \text{CO}_2$.

Sucrose $\xrightarrow{\text{Invertase}}$ Inverted sugar (Glucose + Fructose)



Ans. TGA (Thermal Gravimetric Analysis)	DTA (Diff. Thermal Analysis)	DSC (Differential Scanning Calorimetry)
The change of mass of a sample with change of temp. is observed. ★ Sample can be used as a solid substance.	DTA measure heat diff. b/w a reference sample & a sample of interest over a temp. range. ★ DTA is older ★ DTA can be used at high Temp. ★ Sample can be solid.	DSC measures heat flow of a sample over a temp. range. ★ DSC is modified version of DTA & is new. ★ DSC can not be used. ★ Sample is always a liquid.